

DISCRETE SCHWARZ-PICK LEMMA FOR CIRCLE PACKINGS REVISTED

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ABSTRACT. The Discrete Schwarz-Pick Lemma is a discrete analogue of the classical result from complex analysis, arising from the connection between circle packings and conformal maps established by Thurston. Previous works by Beardon-Stephenson and Van Eeuwen proved this lemma for circle packings where circles are tangent or intersect at non-obtuse angles, corresponding to inversive distances $I \in [0, 1]$. This paper extends the investigation to circle packings with obtuse intersections ($I \in (-1, 0)$) and disjoint packings ($I > 1$). We prove that the Discrete Schwarz-Pick Lemma holds for the full range of intersecting circle packings with inversive distances in $(-1, 1]$, provided an additional condition on the weights of each triangle is satisfied. The proof relies on a variational principle for circle packings with inversive distances. Conversely, we show that the lemma fails for disjoint circle packings where $I \geq 1$. This is demonstrated by constructing a specific counterexample on a triangulated disk with four vertices and providing a numerical analysis to confirm its validity beyond computational error.

1. INTRODUCTION

In March of 1985, in the International Symposium in Celebration of the The connection between circle packings and conformal maps, first suggested by William Thurston, has been a fruitful area of study. Thurston conjectured that the conformal map from a simply connected domain Ω to the unit disk $\mathbb{D}$ could be approximated by circle packings, a claim famously proven by Rodin and Sullivan in 1987 [1]. This bridge between discrete geometry and complex analysis inspired further exploration into discrete analogues of classical theorems.

One such result was the Discrete Schwarz-Pick Lemma. The lemma was first established by Beardon and Stephenson [2] for circle packings where adjacent circles are tangent, corresponding to an inversive distance of $I = 1$. This was later extended by Van Eeuwen to packings with non-obtuse intersections, where $I \in [0, 1]$ [3]. However, it remained an open question whether the lemma would hold for the full range of intersecting circles, which includes obtuse intersections ($I \in (-1, 1)$) or for packings of disjoint circles ($I > 1$). This paper addresses these open cases.

Building on a observation by Zhou [4] and Xu [5], we prove the Discrete Schwarz-Pick Lemma for inversive distances (with a additional condition) between $(-1, 1]$ and show a counter example for inversive distances greater than or equal to 1. Our primary tools include the variational principle for circle packings, introduced by Colin de Verdière [6] and extended by Guo [7], and foundational results on circle packing geometries from Thurston [8].

The paper is organized as follows. Section 2 establishes the main result, proving the Discrete Schwarz-Pick Lemma for intersecting circle packings with inversive distances in $(-1, 1]$. Section 3 constructs a counterexample to show the lemma fails for disjoint packings with inversive distances greater than 1, supported by a numerical analysis to ensure its validity.

2. PRELIMINARIES AND NOTATION

In this subsection, we discuss the concept of inversive distance circle packings introduced by Bowers and Stephenson [9].

Let S be a surface with a finite simplicial triangulation $\mathcal{T}$. We denote the set of vertices, edges, and faces of T by V , E , and F , respectively. Let $V = \{v_1, \dots, v_N\}$ where $N = |V|$.

A weighted simplicial triangulation $(S, \mathcal{T}, I)$ consists of a surface with a simplicial triangulation $\mathcal{T}$ along with a weight function $I : E \rightarrow (-1, \infty)$ that assigns an inversive distance $I_{i,j}$ to each $v_i v_j \in E$. The inversive distance determines the geometry of the circles in the circle packing:

- If $I_{i,j} \in (-1, 1]$ the circles corresponding to vertices v_i and v_j respectively intersect. There exists a unique angle $\Phi_{i,j} \in [0, \pi)$ such that $I_{i,j} = \cos(\Phi_{i,j})$. $\Phi_{i,j}$ is the angle formed by the tangent lines of the circles at the intersection point. See Figure 1 for reference. The intersection is obtuse if $I_{i,j} \in (-1, 0)$, orthogonal if $I_{i,j} = 0$, and acute if $I_{i,j} \in (0, 1)$.
- If $I_{i,j} = 1$, the circles are tangent.
- If $I_{i,j} > 1$, the circles are disjoint.

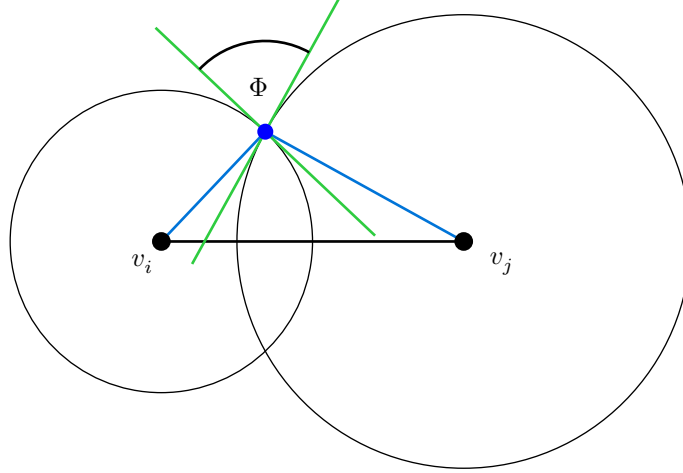


FIGURE 1. Intersection Angle in Circle Packings

Let $\mathbb{R}_{>0} := (0, \infty)$. A radius function is a positive function $r : V \rightarrow \mathbb{R}_{>0}$ where we denote $r_i = r(v_i)$. The radius function and inversive distance induce an edge length function $l_r : E \rightarrow \mathbb{R}_{>0}$ where

$$l_r(v_i v_j) := \sqrt{r_i^2 + r_j^2 + 2r_i r_j I_{i,j}}$$

for Euclidean background geometry and

$$l_r(v_i v_j) := \cosh^{-1}(\cosh(r_i) \cosh(r_j) + I_{i,j} \sinh(r_i) \sinh(r_j))$$

for hyperbolic background geometry. If for every face $\triangle v_i v_j v_k \in F$, if the induced edge lengths l_r satisfy the triangle inequality then the radius function r is called a circle packing metric.

Such a metric induced a discrete curvature $K_r : V \rightarrow (-\infty, 1]$ where

$$K_r(v) := 2\pi - \sum_{\alpha > v} \alpha$$

where $\alpha > v$ means all the angles incident at v .

3. DISCRETE SCHWARZ-PICK LEMMA FOR CIRCLE PACKING WITH INTERSECTIONS

Since this section concerns intersecting circles with inversive distances $I(e) \in (-1, 1]$, it is natural to define the corresponding intersection angle function $\Phi : E \rightarrow [0, \pi)$. For each edge $e \in E$, the angle $\Phi(e)$ is the unique value in $[0, \pi)$ such that $I(e) = \cos(\Phi(e))$.

3.1. Triangle Preliminaries.

Suppose $\triangle v_i v_j v_k \in F$ is a topological triangle. To simplify notation, from here on out, we will use l_i to denote the edge length of the edge $v_j v_k$ and $I_i = I(v_j v_k)$ to denote the inversive distance.

Let

$$\begin{aligned} \gamma_i^{j,k} &= I_i + I_j I_k \\ \gamma_j^{i,k} &= I_j + I_i I_k \\ \gamma_k^{i,j} &= I_k + I_i I_j. \end{aligned}$$

Lemma 3.1.1 ([5]) Suppose $I_i, I_j, I_k \in (-1, 1]$ are three inversive distances where

$$\gamma_i^{j,k}, \gamma_j^{i,k}, \gamma_k^{i,j} \geq 0.$$

For any three positive numbers r_i, r_j, r_k , there exists a configuration of three intersecting disks in Euclidean background geometry and Hyperbolic background geometry, unique up to isometry, having radii r_i, r_j, r_k and with inversive distances of I_i, I_j, I_k .

Lemma 3.1.2 ([5]) For any Euclidean triangle $\triangle v_1 v_2 v_3$, where $\gamma_1^{2,3}, \gamma_2^{1,3}, \gamma_3^{1,2} \geq 0$ and $I : E \rightarrow (-1, 1]$. Let θ_i be the inner angle at vertex v_i and $r_i = e^{u_i}$ for $i \in \{1, 2, 3\}$. Then:

1. $\frac{\partial \theta_i}{\partial u_j} = \frac{\partial \theta_j}{\partial u_i} > 0$ for $i \neq j$.
2. $\frac{\partial \theta_i}{\partial u_i} < 0$.
3. The matrix

$$\left[\frac{\partial \theta_a}{\partial u_b} \right]_{3 \times 3} = \begin{pmatrix} \frac{\partial \theta_1}{\partial u_1} & \frac{\partial \theta_1}{\partial u_2} & \frac{\partial \theta_1}{\partial u_3} \\ \frac{\partial \theta_2}{\partial u_1} & \frac{\partial \theta_2}{\partial u_2} & \frac{\partial \theta_2}{\partial u_3} \\ \frac{\partial \theta_3}{\partial u_1} & \frac{\partial \theta_3}{\partial u_2} & \frac{\partial \theta_3}{\partial u_3} \end{pmatrix}$$

is symmetric and negative semi definite, with null space

$$\mathbb{R}[1, 1, 1] = \{[t, t, t] \mid t \in \mathbb{R}\}.$$

4. The differential 1-form $\sum_{i=1}^3 \theta_i du_i$ is closed on $\mathbb{R}^3$ (and hence exact) and

$$W_{\text{cp}}(u) = \int_0^u \sum_{i=1}^3 \theta_i du_i$$

is a well defined concave function of $u \in \mathbb{R}^3$ satisfying

$$\nabla W_{\text{cp}} = (\theta_1, \theta_2, \theta_3)$$

Furthermore W_{cp} is strictly concave when restricted to $\{tu + (1-t)\hat{u} \mid t \in \mathbb{R}\}$, provided that $u - \hat{u} \neq (a, a, a)$ for $a \in \mathbb{R}$.

Proof Part 1 of the lemma holds by Lemma 2.5 in Xu [5]. Part 2 holds because $\theta_i + \theta_j + \theta_k = \pi$ for $\{i, j, k\} = \{1, 2, 3\}$. Thus

$$\frac{\partial \theta_i}{\partial u_i} + \frac{\partial \theta_j}{\partial u_i} + \frac{\partial \theta_k}{\partial u_i} = 0$$

since $\frac{\partial \theta_i}{\partial u_j} > 0$ for $j \neq i$ by 1, $\frac{\partial \theta_i}{\partial u_i} < 0$. Part 3 holds by Lemma 2.6 in Xu [5]. Finally Part 3 holds by Lemma 2.10 in Xu [5]. $\square$

Lemma 3.1.3 ([5]) For any hyperbolic triangle $\triangle v_1 v_2 v_3$, where $\gamma_1^{2,3}, \gamma_2^{1,3}, \gamma_3^{1,2} \geq 0$ and $I : E \rightarrow (-1, 1]$. Let θ_i be the inner angle at vertex v_i and $u_i = \int_{r_i}^{\infty} \frac{\partial t}{\sinh(t)}$ for $i \in \{1, 2, 3\}$. Then:

1. $\frac{\partial \theta_i}{\partial u_j} = \frac{\partial \theta_j}{\partial u_i} > 0$ for $i \neq j$.
2. $\frac{\partial \theta_i}{\partial u_i} < 0$.
3. The matrix

$$\begin{bmatrix} \frac{\partial \theta_a}{\partial u_b} \end{bmatrix}_{3 \times 3} = \begin{pmatrix} \frac{\partial \theta_1}{\partial u_1} & \frac{\partial \theta_1}{\partial u_2} & \frac{\partial \theta_1}{\partial u_3} \\ \frac{\partial \theta_2}{\partial u_1} & \frac{\partial \theta_2}{\partial u_2} & \frac{\partial \theta_2}{\partial u_3} \\ \frac{\partial \theta_3}{\partial u_1} & \frac{\partial \theta_3}{\partial u_2} & \frac{\partial \theta_3}{\partial u_3} \end{pmatrix}$$

is symmetric and negative semi definite, with null space

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4. The differential 1-form $\sum_{i=1}^3 \theta_i du_i$ is closed on $\mathbb{R}^3$ (and hence exact) and

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is a well defined concave function of $u \in \mathbb{R}^3$ satisfying

$$\nabla W_{\text{cp}} = (\theta_1, \theta_2, \theta_3)$$

Furthermore W_{cp} is strictly concave when restricted to $\{tu + (1-t)\hat{u} \mid t \in \mathbb{R}\}$, provided that $u - \hat{u} \neq (a, a, a)$ for $a \in \mathbb{R}$.

Proof Part 1 of the lemma holds by Lemma 3.6 in Xu [5]. By Remark 11 in Xu [5], $\{i, j, k\} = \{1, 2, 3\}$,

$$\frac{\partial \theta_i}{\partial u_i} + \frac{\partial \theta_j}{\partial u_i} + \frac{\partial \theta_k}{\partial u_i} < 0$$

since $\frac{\partial \theta_i}{\partial u_j} > 0$ for $j \neq i$ by 1, $\frac{\partial \theta_i}{\partial u_i} < 0$, thus Part 2 holds. Part 3 holds by Lemma 3.7 in Xu [5]. Finally Part 3 holds by Lemma 3.9 in Xu [5]. $\square$

3.2. Degeneration of Circle Packing Metrics.

For simplicity, if $U = \{v_j, \dots, v_l\} \subset V$, then we will use I to also denote the index set $\{j, \dots, l\}$.

Lemma 3.2.4 Let $(S, \mathcal{T}, I)$ be a closed connected simplicially-triangulated surface with weight $I : E \rightarrow (-1, 1]$. Let $\Phi : E \rightarrow [0, \pi)$ be the corresponding intersection angle for I , ie $I(e) = \cos(\Phi(e))$. Let U be a proper subset of vertices. Let (r_n) be a sequence of circle packing metrics based on $(S, \mathcal{T}, \Phi)$ in euclidean or hyperbolic background geometry such that $\forall v \in U$ where $\lim_{n \rightarrow \infty} r_n(v) = 0$ and $\forall w \notin U$, $\lim_{n \rightarrow \infty} r_n(w) = 0$. Then

$$\lim_{n \rightarrow \infty} \sum_{v \in U} K_{r_n}(v) = 2\pi\chi(F_U) - \sum_{(e,v) \in \text{Lk}(U)} (\pi - \Phi(e))$$

where F_U is the subcomplex consisting of simplices whose vertices are in U and $\text{Lk}(U) = \{(e, v) \mid e \text{ is an edge so } e \cap U = \emptyset, v \in U \text{ so that } e, v \text{ form a triangle}\}$. Furthermore if $\forall \triangle v_i v_j v_k$ we have that $\gamma_i^{j,k}, \gamma_j^{i,k}, \gamma_k^{i,j} \geq 0$, then for any circle packing metric r based on $(S, \mathcal{T}, \Phi)$ and any proper subset U of vertices V we have,

$$\sum_{u \in U} K_r(u) > 2\pi\chi(F_U) - \sum_{(e,v) \in \text{Lk}(U)} (\pi - \Phi(e))$$

Proof The proof follows the argument outlined in Proposition 4.1 of Chow and Luo [10]. Let $V(\mathcal{T}) = \{v_1, \dots, v_m\}$. Recall θ_i^{jk} is the inner angle at vertex v_i in face $\triangle v_i v_j v_k$. Consider $T \subset F$ where T has a vertex in U . You can partition these triangles into 3 sets, A_1, A_2 , and A_3 where a triangle is in A_i if and only if it has i vertices in U . Thus for any circle packing metric r ,

$$\begin{aligned} \sum_{u \in U} K_r(u) &= 2\pi|I| - \sum_{\substack{\text{all triangles} \\ \triangle v_i v_j v_k \in F}} \theta_i^{jk} \\ &= 2\pi|U| - \left(\sum_{\substack{v_i \in U \\ \triangle v_i v_j v_k \in A_1}} \theta_i^{jk} + \sum_{\substack{v_i, v_j \in U \\ \triangle v_i v_j v_k \in A_2}} (\theta_i^{jk} + \theta_j^{ik}) + \sum_{\triangle v_i v_j v_k \in A_3} (\theta_i^{jk} + \theta_j^{ik} + \theta_k^{ij}) \right) \end{aligned}$$

It is known that $(r_n)_{n \in \mathbb{N}}$ is a sequence of circle packing metrics where $\forall v \in I$, $r_n(v) \rightarrow 0$. Thus the following holds:

1. If $\triangle v_i v_j v_k \in A_3$, as $(r_n(v_i), r_n(v_j), r_n(v_k)) \rightarrow (0, 0, 0)$, $\triangle v_i v_j v_k$ approaches a Euclidean triangle. Thus $\lim_{n \rightarrow \infty} (\theta_i^{jk}(r_n) + \theta_j^{ik}(r_n) + \theta_k^{ij}(r_n)) = \pi$.
2. If $\triangle v_i v_j v_k \in A_2$ and $i, j \in U$, as $(r_n(v_i), r_n(v_j)) \rightarrow (0, 0)$, $\triangle v_i v_j v_k$ approaches a geodesic segment. Thus $\lim_{n \rightarrow \infty} (\theta_i^{jk}(r_n) + \theta_j^{ik}(r_n)) = 0 \Rightarrow \lim_{n \rightarrow \infty} (\theta_i^{jk}(r_n) + \theta_j^{ik}(r_n)) = \lim_{n \rightarrow \infty} (\pi - \theta_k^{ij}(r_n)) = \pi$.
3. If $\triangle v_i v_j v_k \in A_1$ and $i \in U$, then v_i approaches the intersection point of the circles corresponding to v_j and v_k . Thus $\lim_{n \rightarrow \infty} (\theta_i^{jk}(r_n)) = \pi - \Phi(v_j v_k)$. Additionally note $(v_i, v_j v_k) \in \text{Lk}(U)$.

Thus

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{u \in U} K_{r_n}(u) &= 2\pi|U| - \left(\sum_{(v,e) \in \text{Lk}(U)} (\pi - \Phi(e)) + |A_2|\pi + |A_3|\pi \right) \\ &= 2\pi \left(|U| - \frac{|A_2|}{2} - \frac{|A_3|}{2} \right) - \sum_{(v,e) \in \text{Lk}(U)} (\pi - \Phi(e)) \\ &= 2\pi \left(|U| - \frac{1}{2}(|A_2| + 3|A_3|) + |A_3| \right) - \sum_{(v,e) \in \text{Lk}(U)} (\pi - \Phi(e)) \end{aligned}$$

Note $|A_3|$ is the number of 3 cells in F_U , and $|U|$ are the number of 0 cells in F_U . By construction the number of edges in F_U is $\frac{1}{2}(|A_2| + 3|A_3|)$. Thus,

$$\lim_{n \rightarrow \infty} \sum_{u \in U} K_{r_n}(u) = 2\pi\chi(F_U) - \sum_{(v,e) \in \text{Lk}(U)} (\pi - \Phi(e)).$$

For the second part of the proposition, we note that since the background geometry is hyperbolic or euclidean, that the maximum sum of interior angles of a triangle is π . Thus

$$\sum_{\substack{v_i, v_j \in U \\ \triangle v_i v_j v_k \in A_2}} (\theta_i^{jk} + \theta_j^{ik}) < |A_2|\pi, \quad \sum_{\triangle v_i v_j v_k \in A_3} (\theta_i^{jk} + \theta_j^{ik} + \theta_k^{ij}) \leq |A_3|\pi$$

Furthermore, since $\gamma_i^{j,k}, \gamma_j^{i,k}, \gamma_k^{i,j} \geq 0$ for all triangles $\triangle v_i v_j v_k \in F(\mathcal{T})$, by Lemma 3.1.3 and Lemma 3.1.2 for monotonicity, we see that $\pi - \Phi(v_i v_j)$ is the maximal value for θ_i^{jk} for $\triangle v_i v_j v_k \in A_1$ and $v_i \in I$. Thus this shows that $2\pi\chi(F_U) - \sum_{(v_i, e) \in \text{Lk}(U)} (\pi - \Phi(e))$ is strictly less than $\sum_{v \in I} K_r(v)$. $\square$

3.3. Main Theorem.

Theorem 3.3.5 Let $(S, \mathcal{T}, I)$ be a closed connected weighted simplicially-triangulated surface where for any topological triangle $\triangle v_i v_j v_k \in F$ we have $\gamma_i^{j,k} \geq 0, \gamma_j^{i,k} \geq 0, \gamma_k^{i,j} \geq 0$ and $I : E \rightarrow (-1, 1]$. Let Φ be its corresponding intersection angle function. Let $V = V_1 \sqcup V_2$ such that $V_1 \neq \emptyset$ and in the case Euclidean background, $V_2 \neq \emptyset$. Suppose R and r are two circle packing radius assignments on $(S, \mathcal{T})$ with either euclidean or hyperbolic background such that $R|_{V_2} \geq r|_{V_2}$ and their curvatures k_R and k_r satisfy $K_R|_{V_1} \geq K_r|_{V_1}$. Then $R \geq r$.

Proof Let

$$X := \{x \in \mathbb{R}_{>0}^N \mid x|_{V_2} \geq r|_{V_2}, K_x|_{V_1} \geq K_r|_{V_1}\}.$$

Note that $r \in X$ and $R \in X$ by assumption. We want to show that $\forall v \in V, r(v) = \inf(x(v) \mid x \in X)$ or $r = \inf(X)$. This will imply that $R \geq r$ and finish the proof.

First we claim that $\forall a, b \in X$, then $c = \min(a, b) \in X$. Clearly, $c|_{V_2} \geq r|_{V_2}$. $\forall v \in V_1$, let $\{u_1, \dots, u_m\}$ be the set of all vertices adjacent to it. Assume without a loss of generality, $c(v) = a(v)$. Since $c(v_i) \leq a(v_i)$, by part 1 of Lemma 3.1.2 and part 1 of Lemma 3.1.3, we see that $K_c(v) \geq K_a(v) \Rightarrow K_c(v) \geq K_r(v)$. Thus $c \in X$.

Now let $w = \inf(X)$. $\forall v \in V$, let $(x_{v,n})$ be sequence in X such that

$$w(v) = \lim_{n \rightarrow \infty} x_{v,n}(v).$$

Let $(y_n)_{n \in \mathbb{N}}$ where

$$y_n := \min\{x_{v,n} \mid v \in V\}.$$

By the claim proved above, $y_n \in X$ for all $n \in \mathbb{N}$. Since $y_n \leq x_{v,n} \forall v \in V$,

$$\limsup_{n \rightarrow \infty} y_n(v) \leq \limsup_{n \rightarrow \infty} x_{v,n}(v) = \lim_{n \rightarrow \infty} x_{v,n}(v) = w(v).$$

On the other hand, by the definition of infimum, $w \leq y_n$, thus $\forall v \in V$

$$w(v) \leq \liminf_{n \rightarrow \infty} y_n(v).$$

Thus

$$w(v) = \lim_{n \rightarrow \infty} y_n(v).$$

To verify that $w \in X$, we have to verify the three conditions.

1. $w \in \mathbb{R}_{>0}^V$: Assume the contrary that $I = \{v \in V \mid w(v) = 0\} \neq \emptyset$. By **Lemma 3.2.4**, we have

$$\lim_{n \rightarrow \infty} \sum_{v \in I} K_{y_n}(v) = 2\pi\chi(F_I) - \sum_{(e,v) \in \text{Lk}(I)} (\pi - \Phi(e)).$$

Note that $\forall v \in V_2$, $w(v) \geq r(v) > 0 \Rightarrow I \cap V_2 = \emptyset \Rightarrow I \subset V_1$. Thus $\forall v \in I$, $K_{y_n}(v) \geq K_r(v)$. Combining this fact and **Lemma 3.2.4** we see,

$$\lim_{n \rightarrow \infty} \sum_{v \in I} K_{y_n}(v) \geq \sum_{v \in I} K_r(v) > 2\pi\chi(F_I) - \sum_{(e,v) \in \text{Lk}(I)} (\pi - \Phi(e)).$$

This is a contradiction. Thus $I = \emptyset$ and $w \in \mathbb{R}_{>0}^V$ and it is a valid circle packing metric for $(S, \mathcal{T}, \Phi)$.

2. $w|_{V_2} \geq r|_{V_2}$: $\forall n \in \mathbb{N}, y_n|_{V_2} \geq r|_{V_2}$ and since $w = \lim_{n \rightarrow \infty} y_n$. Thus $w|_{V_2} \geq r|_{V_2}$.
3. $K_w|_{V_1} \geq K_r|_{V_1}$: $\forall v \in V_1$,

$$K_w(v) = \lim_{n \rightarrow \infty} K_{r_n}(v)$$

by the sequential definition of continuity.

Thus $w \in X$. By definition of infimum, $r|_{V_2} \geq w|_{V_2}$. Thus $r|_{V_2} = w|_{V_2}$. Assume the contrary $\exists v \in V_1 \ni K_w(v) > K_r(v)$. $\forall t \in (0, w(v))$, let $w^t : V \rightarrow \mathbb{R}$ where

$$w^t(x) := \begin{cases} w(x) - t & \text{if } x = v \\ w(x) & \text{otherwise} \end{cases}.$$

Since $v \in V_1$, $w^t|_{V_2} = w|_{V_2} \geq r|_{V_2}$. Let $\{u_1, \dots, u_m\}$ be all the adjacent vertices to v . Since $w^t(u_i) = w(u_i)$, by part 1 of **Lemma 3.1.2** and part 1 of **Lemma 3.1.3** $K_{w^t}(v) < K_w(v)$ and $K_{w^t}(u_i) \geq K_w(u_i) \geq K_r(u_i)$. By continuity, $\exists t_0 \in (0, w(v))$ small such that $K_r(v) \leq K_{w^{t_0}}(v) \leq K_w(v)$. Thus $w^{t_0} \in X$, however $w^{t_0}(v) < w(v)$. This is a contradiction since w is a infimum. Thus $K_w|_{V_1} = K_r|_{V_1}$.

Now we have to show that $r = w$ given the fact that $r|_{V_2} = w|_{V_2}$ and $K_w|_{V_1} = K_r|_{V_1}$. Let $A = \mathbb{R}$ if the background geometry is Euclidean and $A = (0, \infty)$ if the background geometry is Hyperbolic. Let $\varphi : (0, \infty) \rightarrow A$ where

$$\varphi(x) := \begin{cases} \ln(x) & \text{if Euclidean Background} \\ \int_x^\infty \frac{dt}{\sinh(t)} & \text{if Hyperbolic Background} \end{cases}$$

φ is a homeomorphism so φ^{-1} exists and is a continous. Let $\mathcal{R} = \{x \in \mathbb{R}_{>0}^V \mid \varphi^{-1}(x)|_{V_2} = r|_{V_2}\}$. Let $W : A^V \rightarrow \mathbb{R}$ where

$$W(x) := \sum_{\triangle v_i v_j v_k \in F} W_{\text{cp}}(x_i, x_j, x_k).$$

part 1 of **Lemma 3.1.2** and part 1 of **Lemma 3.1.3**,

$$\begin{aligned}
 \frac{\partial W}{\partial x_a} &= \sum_{\triangle v_a v_j v_k \in F} \frac{\partial W_{\text{cp}}}{\partial x_a} \\
 &= \sum_{\triangle v_a v_j v_k \in F} \theta_a^{j,k} \\
 &= 2\pi - K_{\varphi^{-1}(x)}(v_a).
 \end{aligned}$$

Thus

$$\nabla W(x) = 2\pi - K_{\varphi^{-1}(x)}.$$

Let $F = W|_{\mathcal{R}}$. Thus $F : \mathbb{R}_{>0}^{V_1} \rightarrow \mathbb{R}$ and

$$\nabla F = 2\pi - K_{\varphi^{-1}(x)}|_{V_1}.$$

$\forall x, y \in \mathcal{R}$ such that $x \neq y$. Take $M = \{v \in V \mid x(v) \neq y(v)\}$ and $N = \{v \in V \mid x(v) = y(v)\}$. Note $M \neq \emptyset$ since $x \neq y$, $V_2 \subset N$, and $M \cap N = \emptyset$. Since $(S, \mathcal{T})$ is connected, then the graph of the triangulation is connected. Thus $\exists (a, b) \in M \times N$ such that $\exists \triangle abc \in F$.

$$(x(a) - y(a), x(b) - y(b), x(c) - y(c)) = (x(a) - y(a), 0, x(c) - y(c)) \neq (0, 0, 0)$$

since $x(a) \neq y(a)$. Thus W_{cp} is strictly concave on $\{tu + (1-t)\hat{u}\}$ where $u = (x(a), x(b), x(c))$ and $\hat{u} = (y(a), y(b), y(c))$. Thus W_{cp} is strictly concave on $\mathcal{R}$, thus W and F are strictly concave on $\mathcal{R}$. Thus ∇F is injective. Thus $x \rightarrow K_{\varphi^{-1}(x)}|_{V_1}$ is injective. Thus since $K_w|_{V_1} = K_r|_{V_1} \Rightarrow w|_{V_1} = r|_{V_1} \Rightarrow w = r$. Thus $r = \inf(X)$ and $R \geq r$. □

4. DISCRETE SCHWARZ-PICK LEMMA FOR $I \geq 1$

For $I \geq 1$, we conjectured that the statement of the Discrete Schwarz would hold because for such circle packings we have rigidity by Guo [7] and Luo [11].

Conjecture 4.6 Let $(S, \mathcal{T}, I)$ be a closed connected weighted simplicially-triangulated surface where for any topological triangle $\triangle v_i v_j v_k \in F$ we have $\gamma_i^{j,k} \geq 0, \gamma_j^{i,k} \geq 0, \gamma_k^{i,j} \geq 0$ and $I \geq 1$. Let $V = V_1 \sqcup V_2$ such that $V_1 \neq \emptyset$ and in the case Euclidean background, $V_2 \neq \emptyset$. Suppose R and r are two circle packing radius assignments on $(S, \mathcal{T})$ with either euclidean or hyperbolic background such that $R|_{V_2} \geq r|_{V_2}$ and their curvatures k_R and k_r satisfy $k_R|_{V_1} \geq k_r|_{V_1}$. Then $R \geq r$.

4.1. Counterexample for $I \geq 1$ on the disk with boundary.

The conjecture concerns closed surfaces, but a counterexample on a triangulated disk with a boundary is sufficient. By a standard doubling argument, any such counterexample on a disk can be extended to a closed surface.

Let $(S, \mathcal{T}, I)$ be a disk with boundary with interior vertices V_1 and boundary vertices V_2 . Construct a closed surface S' by taking two copies of S and identifying their boundaries with the identity map. This induces a triangulation T' and inverse distance function I' on S' . Given two metrics r and R on $(S, \mathcal{T}, I)$, we define corresponding metrics r' and R' on $(S', \mathcal{T}', I')$. If the premises of the conjecture (ie $r(v) \leq R(v)$ for all $v \in V_2$ and $K_r(v) \leq K_R(v)$ for all $v \in V_1$) hold on the disk, they will also hold on the doubled surface. Therefore if we find an interior vertex $d \in V_1$

such that $r(d) > R(d)$, this violation will carry over to the corresponding vertices in the doubled surface S' and thus refuting the conjecture for closed surfaces.

4.2. Construction of the Counterexample.

We now construct such a disk:

- **Vertices:** The vertex set is $V = \{a, b, c, d\}$.
- **Partition:** We partition the vertices as $V_1 = \{d\}$ and $V_2 = \{a, b, c\}$.
- **Triangulation:** The surface consists of three triangles: (a, b, d) , (a, c, d) , and (b, c, d) .
- **Inversive Distance Edge Weights:** The weights I are given as follows:

Edge	Inversive Distance (I)
ab	1.0
ac	1.0
bc	1.0
ad	1.19301
bd	4.14177
cd	3.14727

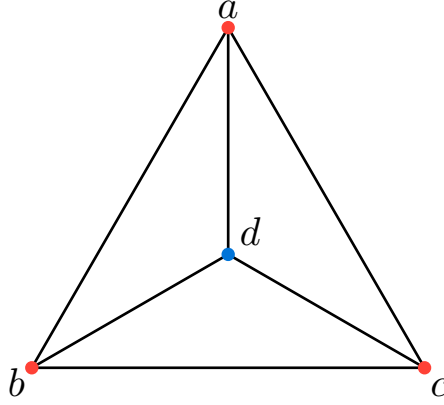


FIGURE 2. Triangulation of Counter Example

We now define two circle packing metrics, r and R , on this surface. The radii and resulting discrete curvatures for the metric r are:

Vertex	Radius $r(v)$	Curvature $k_r(v)$
a	100.0	2.49300
b	100.0	4.54271
c	100.0	3.97976
d	148.29698	4.69250

The induced edge lengths from this circle packing metric are:

Edge e	Edge Length $l_r(e)$
ab	200.0
ac	200.0
bc	200.0
ad	259.56903
bd	393.49002
cd	354.03124

The radii and resulting discrete curvatures for the metric R are:

Vertex	Radius $R(v)$	Curvature $k_R(v)$
a	109.19267	1.24905
b	245.79886	5.20030
c	222.06186	4.49970
d	147.98050	4.75891

The induced edge lengths from this circle packing metric are:

Edge e	Edge Length $l_R(e)$
ab	354.99153
ac	331.25453
bc	467.86072
ad	269.02721
bd	619.36753
cd	527.30776

4.3. Verification of the Counterexample.

We now check the conditions of the conjecture.

4.3.1. Checking Initial Assumptions.

1. **Condition on V_2 :** We must check if $r(v) < R(v)$ for all $v \in V_2 = \{a, b, c\}$.

- $r(a) = 100.0 < 109.19267 = R(a)$
- $r(b) = 100.0 < 245.79886 = R(b)$
- $r(c) = 100.0 < 222.06186 = R(c)$

The condition holds.

2. **Condition on V_1 :** We must check if $k_r(v) \leq k_R(v)$ for all $v \in V_1 = \{d\}$.

- $k_r(d) = 4.69250 \leq 4.75891 = k_R(d)$

This condition also holds.

Both premises of the conjecture are satisfied.

4.3.2. Violation of the Conclusion.

The conjecture concludes that $r(v) \leq R(v)$ for **all** vertices $v \in V$. We have already verified this for V_2 . We now check the vertex in V_1 :

- For vertex d :

- $r(d) = 148.29698$
- $R(d) = 147.98050$

Here, we find that $r(d) > R(d)$.

4.4. Analysis of the Effects of Numerical Precision.

This counterexample was generated computationally, which necessitates the analysis of the role of floating point arithmetic. We must ensure that the key result, the violation of $r(d) > R(d)$, is not a artifact of numerical error.

The core of the counter example is the inequality $r(d) > R(d)$. The difference is $r(d) - R(d) \approx 0.00316$. The computations were performed with standard IEEE 754 double precision arithmetic, which has a machine epsilon of $\varepsilon_{\text{mach}} = 2.22 \times 10^{-16}$. This value represents the smallest number that when added to 1, gives a different result than 1. The observed difference is $\approx 3.16 \times 10^{-3}$ which is 13 orders of magnitude greater than machine epsilon, thus making it highly unlikely to be a result of random numerical noise

The calculation of discrete curvature involves 4 main steps: calculation of edge lengths, verification of triangle inequality, finding corner angles using cosine law, and summing these angles.

The formula to calculate edge lengths $l(vw)^2 = r(v)^2 + r(w)^2 + 2r(v)r(w)I(vw)$ involves multiplications and additions. For well scaled inputs, the relative error of these operations is on the order of $\varepsilon_{\text{mach}}$. The subsequent square root is a well condition operation that keeps the relative error on the order of $\varepsilon_{\text{mach}}$. Thus the relative error is on the order of $\varepsilon_{\text{mach}}$ to find edge lengths.

To verify the triangle inequality, for each triangle $\triangle v_i v_j v_k$ we do the checks $l(v_i v_j) + l(v_i v_k) > l(v_j v_k)$, $l(v_i v_j) + l(v_j v_k) > l(v_i v_k)$, and $l(v_i v_k) + l(v_j v_k) > l(v_i v_j)$ to verify the triangle inequality. This is a sequence of additions which contribute a relative error on the order of $\varepsilon_{\text{mach}}$. The check fails at less then the order of $\varepsilon_{\text{mach}}$. The lengths of the edges, such that any failure in the triangle inequality doesn't occur.

To calculate the interior angles we use the law of cosines, $\theta = \arccos\left(\frac{a^2+b^2-c^2}{2ab}\right)$, which is the the most sensitive step. It can suffer from catastrophic cancellation, if the triangle is nearly degenerate. This would cause a significant loss in precision. However, an analysis of the induced edge lengths for both metrics r and R shows that the triangles are well-conditioned and far from degenerate. Therefore, we do not expect a dramatic loss of precision in the angle calculations.

The final curvature $k(v) = 2\pi - \sum_{\alpha > v} \alpha_i$. At each vertex, k_v is the sum of at most 3 angles. The total error from each angle remains small.

In summary, while all floating-point computations have inherent error, the structure of the calculations in this problem is numerically stable. The magnitude of the violation $r(d) > R(d)$ is vastly larger than the expected accumulated error. The observed result is therefore a robust feature of this specific geometric configuration and serves as a valid refutation of the conjecture.

4.5. Conclusion of Counterexample.

The premises hold ($r|_{V_2} \leq R|_{V_2}$, $k_r|_{V_1} \leq k_R|_{V_1}$), but $r(d) > R(d)$. This violates the conjecture on the disk. By the doubling argument, it also provides a counterexample for closed surfaces.

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